Network-Conditioned Factor Predictability from Co-Trading Synchronization

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Abstract

This paper studies whether ETF returns predict cross-sectional stock returns in a way that depends on market-wide co-trading synchronization. The benchmark design regresses individual stock close-to-close returns on stock-level heterogeneous autoregressive controls, ETF close-to-close predictors, and interactions between ETF predictors and a high-synchronization regime constructed from abnormal stock-level volume across the full equity universe. The main finding is sharply state-dependent. For broad equity ETFs, daily predictability is weakly positive or near zero in low-synchronization periods but turns strongly negative in high-synchronization periods, while weekly and monthly predictability become more positive. The paper then develops two extensions. First, a return-timing decomposition separates close-to-open and open-to-close components on both the predictor and target sides, showing that daily reversal appears in both components, weekly continuation is concentrated in intraday ETF moves, and monthly continuation is concentrated in overnight ETF moves. Second, ETF-matched synchronization measures are built from economically aligned stock universes rather than the full market. These matched regimes are meaningfully different from the benchmark synchronization calendar, especially for sector ETFs, though they do not uniformly dominate the market-wide measure. Finally, out-of-sample forecasting and portfolio evidence show that ETF-augmented models improve cross-sectional ranking performance and gross portfolio quality more than they improve point-forecast accuracy, while turnover remains the main obstacle to net profitability. Overall, the evidence supports a regime-dependent view of ETF-based predictability in which synchronization governs whether predictability exists, what sign it takes, and when it appears.

Keywords

ETF predictability, synchronization, heterogeneous autoregressive models, overnight returns, intraday returns, return forecasting

1 Introduction

Exchange-traded funds (ETFs) aggregate information, liquidity demand, hedging pressure, and delegated portfolio flows into highly liquid market objects. If ETF returns forecast subsequent stock returns, the predictive relation can plausibly reflect several mechanisms at once: information transmission, temporary price pressure, sentiment, or coordinated trading demand. A natural question is whether ETF-based predictability is stable, or whether its sign and timing depend on the market state.

This paper studies that question through the lens of *co-trading synchronization*. Intuitively, if abnormal volume shocks occur across many stocks at the same time, the market may be experiencing broad coordinated trading pressure rather than isolated stock-specific news. In such an environment, ETF moves may forecast subsequent stock returns differently than in ordinary periods. A signal that looks like continuation in a low-synchronization environment may become reversal in a high-synchronization environment if contemporaneous ETF moves partly reflect temporary crowding or overshooting.

The paper is organized around one benchmark empirical design and two extensions. The benchmark specification is the core contribution: stock close-to-close returns are regressed on stock-level heterogeneous autoregressive (HAR) controls, ETF close-to-close predictors, and interactions between ETF predictors and a market-wide synchronization indicator constructed from the full stock universe. Extension 1 asks *when* the effect appears by decomposing both predictors and targets into overnight and intraday components. Extension 2 asks whether the synchronization state should be measured market-wide or on ETF-matched stock universes such as large-cap, small-cap, or sector-linked subsets.

The benchmark results are clear. For broad equity ETFs such as SPY, QQQ, and IWM, low-synchronization periods exhibit weak daily continuation or near-zero daily predictability, whereas high-synchronization periods exhibit strong daily reversal. At slower horizons, weekly and especially monthly effects become more positive in high-synchronization regimes. This conditional structure is difficult to reconcile with a single unconditional interpretation of ETF predictability. Instead, it suggests that the same ETF move reflects different economic forces depending on whether the market is undergoing broad synchronized trading.

The timing decomposition provides a sharper mechanism interpretation. Daily reversal appears in both the overnight and intraday components, but weekly continuation is concentrated primarily in intraday ETF signals while monthly continuation is concentrated primarily in overnight ETF signals. A useful summary is that intraday ETF movements appear more closely tied to trading pressure, while overnight ETF movements appear more closely tied to information flow or slower-moving valuation updates.

The ETF-matched synchronization extension is more nuanced. Synchronization calendars built from ETF-relevant stock universes differ materially from the market-wide benchmark, especially for narrower sector universes. However, matched synchronization does not uniformly strengthen the benchmark results. In the current sample, market-wide synchronization remains the strongest state variable for the benchmark daily reversal result, while matched synchronization functions primarily as a diagnostic of economic alignment and a possible refinement for slower-horizon sector effects.

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Finally, the paper evaluates whether the in-sample evidence translates into economically meaningful predictive content. Out-of-sample forecasting results show that ETF-augmented models generally improve cross-sectional ranking metrics relative to HAR-only baselines for several major ETFs. Simple long-short portfolio tests also improve in gross terms when ETF information is included. However, once turnover-based transaction costs are imposed, net performance becomes negative across specifications, indicating that the current implementation remains too friction-intensive to support a strong profitability claim without further refinement.

This paper makes four contributions. First, it documents a benchmark state-dependent sign flip in ETF-based stock return predictability. Second, it shows that the benchmark effect is timing-dependent, not merely state-dependent. Third, it demonstrates that ETF-matched synchronization regimes are economically distinct from the market-wide benchmark even when they do not dominate it. Fourth, it provides secondary out-of-sample and portfolio evidence that the predictive signal has meaningful ranking content and gross economic value, while also identifying turnover as the main implementation constraint.

2 Related Literature

This project connects to several strands of empirical finance.

First, it relates to the literature on return continuation and reversal. Momentum and contrarian effects are among the most studied cross-sectional patterns in asset pricing, with classic evidence documented by Jegadeesh and Titman [6] and Lo and MacKinlay [7]. The present results contribute a conditional perspective: ETF-linked predictability is not simply momentum or simply reversal, but changes sign depending on the synchronization regime.

Second, the paper connects to models of investor sentiment, noise trading, and price pressure. De Long et al. [5] and Barberis et al. [1] emphasize that coordinated demand shocks can move prices away from fundamentals. The synchronization measure used here is designed to proxy for broad simultaneous trading intensity, making the high-synchronization reversal result naturally consistent with temporary pressure or overreaction.

Third, the analysis relates to delegated trading, forced sales, and flow propagation. Coval and Stafford [4] show that common ownership and institutional trading pressure can generate predictable patterns across securities. ETFs are a particularly plausible transmission channel because they aggregate broad exposures into tradeable baskets and can concentrate common flows.

Fourth, the paper is linked to work on ETF market structure and spillovers. Ben-David et al. [2] document that ETFs affect volatility behavior in underlying securities. My results complement that literature by showing that ETF returns can also have predictive content for subsequent stock returns, and that this predictive content is sharply state-dependent.

Finally, the use of daily, weekly, and monthly lags is motivated by the multi-horizon logic of heterogeneous autoregressive models [3]. Although HAR was originally proposed for realized volatility, its layered horizon structure is well suited to organizing short-, intermediate-, and longer-horizon return dynamics in the present setting.

3 Data and Return Construction

3.1 Sample and Universe

The empirical sample spans 2000 through 2023 and is built from a fully reconstructed daily pipeline. The project uses stock-level and ETF-level adjusted prices and volumes. A central design principle is that return series are constructed directly from adjusted prices rather than relying on precomputed vendor return fields. This avoids inconsistencies across sources and ensures internal coherence across close, open, and decomposition-based return definitions.

The stock universe serves two purposes. First, it provides the cross section for pooled stock-level regressions. Second, it provides the underlying panel from which synchronization states are constructed. Because ETF regressors are common across stocks on a given day, the time dimension is the main source of identifying variation for ETF-side coefficients even though the regressions are estimated in pooled form.

3.2 Return Definitions

All returns are constructed from adjusted prices. Let C_t denote the adjusted close on day t and O_t denote the adjusted open on day t. Then

$$r_t^{cc} = \log\left(\frac{C_t}{C_{t-1}}\right), \tag{1}$$

$$r_t^{co} = \log\left(\frac{O_t}{C_{t-1}}\right), \tag{2}$$

$$r_t^{oc} = \log\left(\frac{C_t}{O_t}\right). \tag{3}$$

By construction,

$$r_t^{cc} = r_t^{co} + r_t^{oc}. \tag{4}$$

These definitions are applied consistently to both stocks and ETFs. The benchmark specification uses close-to-close returns on both sides. Extension 1 then decomposes both predictors and targets into overnight and intraday components.

3.3 HAR Aggregation

To organize return persistence across horizons, I use daily, weekly, and monthly predictors. For a generic return series x_t, define

$$x_t^{(d)} = x_t, \tag{5}$$

$$x_t^{(w)} = \frac{1}{5}\sum_{k=1}^{5} x_{t-k}, \tag{6}$$

$$x_t^{(m)} = \frac{1}{22}\sum_{k=1}^{22} x_{t-k}. \tag{7}$$

These transformed series are built separately for stock returns and ETF returns, and separately for close-to-close, close-to-open, and open-to-close returns when needed.

4 Synchronization Measure

The synchronization state variable is constructed from stock-level abnormal volume. Let $V_{i,t}$ denote volume for stock i on day t. For

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each stock, I standardize volume using a rolling mean and standard deviation:

$$z_{i,t} = \frac{V_{i,t} - \mu_{i,t}}{\sigma_{i,t}}. \tag{8}$$

To focus on unusually large shocks, I threshold the standardized series:

$$\tilde{z}_{i,t} = z_{i,t}\mathbf{1}\{|z_{i,t}| > \tau\}. \tag{9}$$

The scalar synchronization index is the cross-sectional sum of squared thresholded shocks:

$$S_t = \sum_i \tilde{z}_{i,t}^2. \tag{10}$$

Days are classified as high synchronization when S_t falls in the top 20% of its empirical distribution:

$$\text{HighSync}_t = \mathbf{1}\{S_t \geq q_{0.8}(S)\}. \tag{11}$$

The purpose of this construction is not simply to proxy for volatility. Rather, it is designed to capture broad simultaneous trading intensity across many stocks, which is more naturally associated with crowding, common flow shocks, or market-wide coordination. In the benchmark specification, the state variable is computed using the full stock universe. In Extension 2, the same construction is applied to ETF-matched stock subsets.

5 Benchmark Empirical Design

The benchmark specification studies whether ETF close-to-close returns predict stock close-to-close returns conditional on market-wide synchronization. For stock i and ETF j, the regression is

$$\begin{aligned} r_{i,t}^{cc} = \alpha &+ \sum_{s\in\{d,w,m\}} \beta^s r_{i,t-1}^{cc,s} + \sum_{s\in\{d,w,m\}} \theta_j^s f_{j,t-1}^{cc,s} \\ &+ \sum_{s\in\{d,w,m\}} \delta_j^s \left(f_{j,t-1}^{cc,s} \cdot \text{HighSync}_{t-1} \right) + \varepsilon_{i,t}. \end{aligned} \tag{12}$$

Here $r_{i,t-1}^{cc,s}$ denotes stock-side HAR controls and $f_{j,t-1}^{cc,s}$ denotes ETF-side daily, weekly, and monthly close-to-close predictors. The coefficients θ_j^s measure ETF effects in low-synchronization states, while the interaction terms δ_j^s measure how those effects change in high-synchronization periods.

An important econometric caveat is that ETF regressors are identical across stocks at each date. As a result, pooled OLS can overstate statistical precision for ETF-side coefficients because the cross-sectional dimension expands the number of observations without creating independent variation in the common ETF regressors themselves. Accordingly, the main objects of interest in this paper are coefficient signs, relative magnitudes, cross-ETF patterns, and stability across specifications rather than nominal pooled OLS t-statistics.

6 Main Benchmark Results

The benchmark close-to-close results are summarized in Figure 1. This figure is the central empirical result of the paper.

The pattern is strongest for broad equity ETFs and sector ETFs linked to equities. In low-synchronization periods, daily ETF effects are weakly positive or near zero. In high-synchronization periods, daily effects become strongly negative. For example, SPY's daily effect moves from approximately 0.024 in low synchronization to

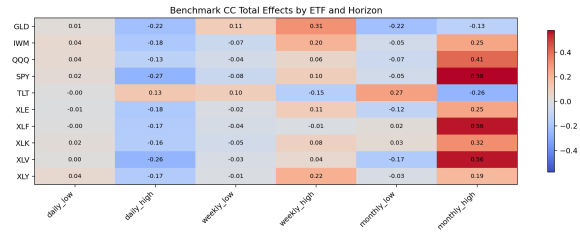

Figure 1: Benchmark total ETF effects in low- and high-synchronization states. Rows correspond to ETFs and columns correspond to total daily, weekly, and monthly effects under low and high synchronization.

−0.267 in high synchronization, QQQ moves from roughly 0.035 to −0.130, and IWM moves from roughly 0.041 to −0.184.

This sign flip is the benchmark contribution of the project: high synchronization turns short-horizon equity ETF predictability from weak continuation into pronounced reversal. A natural interpretation is that in high-synchronization states, ETF moves are more heavily contaminated by crowding or transitory pressure, so next-period stock returns partially unwind the initial ETF-linked move.

At slower horizons, the pattern changes. Weekly and especially monthly effects become more positive in high synchronization. For SPY, the monthly total effect rises from roughly −0.049 in low synchronization to about 0.577 in high synchronization. Similar monthly strengthening appears for QQQ, IWM, XLF, and XLV. This suggests that high-synchronization environments are associated not only with short-horizon overshooting, but also with stronger medium-horizon continuation or gradual information transmission.

An important exception is TLT. Unlike the equity ETFs, TLT often becomes more predictive rather than less predictive in high synchronization states. This is consistent with a different mechanism, potentially related to macro information or flight-to-quality dynamics, rather than the equity crowding channel that dominates the benchmark results for SPY, QQQ, and sector equity ETFs.

7 Extension 1: Timing Decomposition

The benchmark close-to-close results show that synchronization matters, but they do not reveal *when* the effect arises. A close-to-close return conflates overnight price adjustment with intraday trading dynamics. To distinguish these channels, the first extension decomposes both predictor and target returns into close-to-open and open-to-close components.

The fully decomposed specification allows both stock-side controls and ETF-side predictors to enter separately by timing:

$$\begin{aligned} y_{i,t} = \alpha &+ \sum_{s\in\{d,w,m\}} \beta^{co,s} r_{i,t-1}^{co,s} + \sum_{s\in\{d,w,m\}} \beta^{oc,s} r_{i,t-1}^{oc,s} \\ &+ \sum_{s\in\{d,w,m\}} \theta_j^{co,s} f_{j,t-1}^{co,s} + \sum_{s\in\{d,w,m\}} \theta_j^{oc,s} f_{j,t-1}^{oc,s} \\ &+ \sum_{s\in\{d,w,m\}} \delta_j^{co,s} \left(f_{j,t-1}^{co,s} \cdot \text{HighSync}_{t-1} \right) \\ &+ \sum_{s\in\{d,w,m\}} \delta_j^{oc,s} \left(f_{j,t-1}^{oc,s} \cdot \text{HighSync}_{t-1} \right) + \varepsilon_{i,t}, \end{aligned} \tag{13}$$

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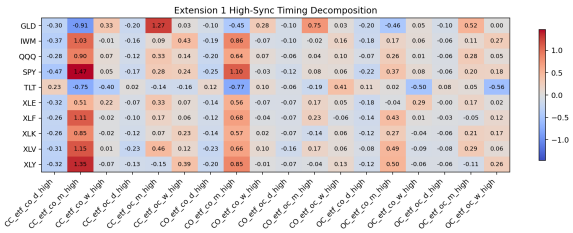

Figure 2: High-synchronization timing decomposition of ETF effects. Columns split ETF predictors into overnight and intraday components across daily, weekly, and monthly horizons, and the rows correspond to ETFs.

where the target $y_{i,t}$ can be close-to-close, close-to-open, or open-to-close.

The decomposition results are summarized in Figure 2. Three findings are especially clear.

First, daily reversal in high-synchronization states appears in both overnight and intraday ETF components. The short-horizon reversal result is therefore not confined to a single segment of the trading day.

Second, weekly continuation is primarily intraday. For SPY, IWM, XLK, and XLY, the high-synchronization weekly effect is much stronger for ETF open-to-close predictors than for ETF close-to-open predictors. This suggests that medium-horizon continuation is closely linked to intraday trading pressure or same-day capital allocation behavior.

Third, monthly continuation is primarily overnight. For SPY, IWM, QQQ, XLF, and XLV, the high-synchronization monthly effect is much larger for ETF close-to-open predictors than for ETF open-to-close predictors. This points to a different mechanism at longer horizons, one that is more consistent with overnight information flow, macro repricing, or gradual valuation adjustment.

Taken together, Extension 1 is the most direct mechanism result in the paper: daily reversal appears in both components, weekly continuation is primarily intraday, and monthly continuation is primarily overnight.

8 Extension 2: Matched Synchronization Universes

The benchmark synchronization measure is built from the full stock universe. This is conceptually appealing because it captures aggregate market-wide crowding, but it may be too broad if one wants the state variable to be tightly aligned with the ETF driving predictability. Extension 2 therefore replaces the market-wide synchronization calendar with ETF-specific matched calendars constructed from economically aligned stock universes.

The matched-universe mapping uses broad size and sector logic. SPY is linked to a large-cap stock universe, IWM to a small-cap stock universe, and sector ETFs to SIC-based subsets of the stock universe. For each ETF-linked universe, the same abnormal-volume synchronization construction is recomputed.

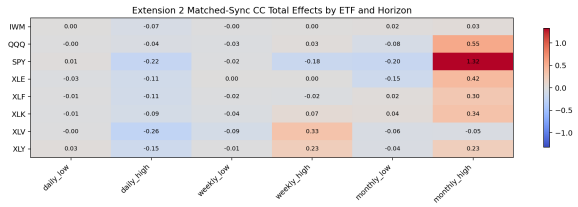

Figure 3: Extension 2 matched-synchronization total ETF effects by horizon. This figure parallels the benchmark heatmap but replaces the market-wide synchronization indicator with ETF-specific matched synchronization regimes.

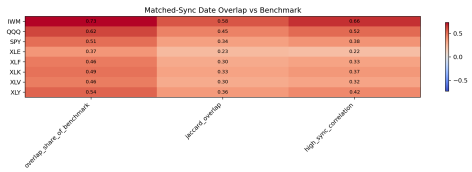

Figure 4: Overlap diagnostics between ETF-matched synchronization calendars and the benchmark market-wide synchronization measure.

Figure 3 shows that the broad reversal story survives the move from market-wide synchronization to ETF-specific synchronization: daily high-synchronization effects remain negative across the matched equity ETFs. However, the magnitude of reversal is often smaller than in the benchmark specification, especially for IWM, QQQ, XLE, and XLF. This suggests that the strongest short-horizon reversal channel remains tied to broad market-wide crowding rather than purely local ETF-specific crowding.

Matched synchronization sometimes sharpens slower-horizon effects. SPY's matched monthly high effect is substantially larger than its market-wide counterpart, and XLE also shows stronger matched monthly continuation. This suggests that ETF-specific synchronization may be more informative for slower-horizon sector or style propagation than for the benchmark daily reversal effect.

Figure 4 shows that matched synchronization is not a trivial relabeling of the benchmark market-wide state. IWM has an overlap share of approximately 0.732 and a high-sync correlation of about 0.664 with the benchmark, while XLE has an overlap share of only about 0.373 and a correlation near 0.216. Broad universes therefore track the market-wide benchmark more closely, while narrower sector universes produce genuinely different synchronization calendars.

The proper conclusion is not that matched synchronization replaces the benchmark. Rather, it functions as a disciplined extension. The benchmark captures broad market-wide crowding; the matched version tests whether ETF-specific synchronization adds structure beyond that market-wide channel.

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9 Secondary Evidence: Out-of-Sample Forecasting

To evaluate whether the in-sample regression evidence translates into out-of-sample predictive content, I compare four expanding-window forecasting models: HAR only, ETF HAR, synchronization-conditioned ETF HAR, and matched-synchronization ETF HAR.

The most relevant out-of-sample metric is the *daily information coefficient*, reported as `daily_ic_mean` in the pipeline output. This metric is computed as the cross-sectional Pearson correlation, on each day, between predicted stock returns and realized stock returns, then averaged across days. In other words, it measures ranking quality more than exact point-forecast accuracy.

Table 1 shows that ETF-augmented models improve out-of-sample ranking performance for several major ETFs. The gains are strongest for XLY, IWM, XLE, XLF, and SPY. An important nuance is that the best-performing specification differs across ETFs: synchronization-conditioned models dominate for several broad equity ETFs, while matched-synchronization models perform best for XLE and XLF. This is consistent with the in-sample evidence that matched synchronization is most useful as a targeted extension rather than as a universal replacement for the benchmark.

By contrast, out-of-sample RMSE does not improve in a meaningful way. Table 2 shows that HAR-only remains the best RMSE model for every ETF after rounding. This distinction is important. The ETF and synchronization variables improve the *ordering* of stocks more than they improve exact level prediction.

The correct interpretation of the forecasting evidence is therefore: better ranking, not better RMSE. That is still economically meaningful in a cross-sectional setting, because portfolio construction typically uses the signal as a sorter rather than as an exact return-level forecast.

10 Secondary Evidence: Portfolio Results

To evaluate economic significance, I map the forecast signals into simple long-short portfolios and compute summary statistics. The portfolio exercise is intentionally simple and should be interpreted as a first pass rather than a production strategy.

I focus first on gross performance, because the gross results isolate whether the signal has economic content before turnover frictions are imposed.

Table 3 shows that ETF-based models improve gross portfolio quality for nearly every ETF. The strongest gross gains appear in XLE, XLF, SPY, IWM, and XLY. These results are consistent with the out-of-sample ranking evidence: the predictive signal contains economically meaningful sorting information when evaluated through stock selection rather than through exact point forecasts.

However, the earlier cost-sensitive portfolio analysis also shows that turnover is a major implementation constraint. Once turnover-based trading costs are imposed, net Sharpe becomes negative across specifications. This does *not* imply that the signal is absent. Rather, it implies that the current portfolio mapping is too friction-intensive. A more mature implementation would likely require signal smoothing, slower rebalancing, threshold-based trading, or more selective portfolio formation.

The portfolio evidence therefore supports a careful conclusion. Gross of trading frictions, the ETF and synchronization variables carry economically meaningful sorting information. Net of realistic implementation costs, the current version is not yet trading-ready.

11 Limitations and Future Work

Several limitations are important.

First, the pooled OLS design requires caution in interpreting nominal significance levels for ETF-side coefficients. Because ETF predictors are common across stocks at each date, the cross-sectional dimension inflates the raw observation count without generating independent ETF-side variation. For this reason, the main empirical emphasis is on signs, relative magnitudes, and cross-specification stability.

Second, synchronization is a reduced-form state variable. It is plausibly related to crowding and common trading intensity, but it does not directly observe investor intent, order-flow direction, or liquidity demand.

Third, the matched synchronization extension uses economically motivated approximations to ETF-linked universes rather than exact time-varying ETF constituent histories. A richer implementation could use holdings-level data or exact constituent membership to refine the state construction.

Fourth, the current forecasting and portfolio exercises are intentionally simple. The negative net Sharpe results do not negate the predictive content of the signal, but they do show that implementation details matter materially.

These limitations point naturally to future work. The most ambitious next step would be to replace the scalar synchronization index with a *pairwise co-trading matrix*. A pairwise measure would capture which stocks are synchronizing with which other stocks, not merely the overall intensity of aggregate synchronization. That would allow the project to study network topology, sector clustering, and market leadership directly. Other extensions include richer ETF holdings-based matching, rank-based information coefficients such as Spearman IC, lower-turnover portfolio design, alternative state variables such as volatility or liquidity regimes, and broader robustness analysis over threshold and rolling-window choices.

12 Conclusion

This paper studies ETF-based stock return predictability through the lens of co-trading synchronization. The main result is that ETF predictability is strongly state-dependent. In low-synchronization periods, broad equity ETFs exhibit weak daily continuation or near-zero effects. In high-synchronization periods, the daily effect flips to pronounced reversal. At the same time, weekly and monthly continuation strengthen in high-synchronization states.

The timing decomposition shows that the benchmark result is not only state-dependent but also timing-dependent. Daily reversal appears in both overnight and intraday components, weekly continuation appears mostly intraday, and monthly continuation appears mostly overnight. This pattern supports an interpretation in which intraday ETF moves are more tightly linked to trading pressure, while overnight ETF moves are more closely linked to information flow.

The matched synchronization extension shows that synchronization calendars can differ meaningfully across economically

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Table 1: Out-of-sample ranking summary by ETF. The final column reports the gain in daily information coefficient relative to the HAR-only baseline.

ETF	HAR-only IC	Best model	Best IC	IC gain vs. HAR
XLY	0.002747	sync_etf_har	0.004541	0.001794
IWM	0.002349	sync_etf_har	0.003763	0.001413
XLE	0.002747	matched_sync_etf_har	0.003856	0.001109
XLF	0.002747	matched_sync_etf_har	0.003504	0.000758
SPY	0.002747	sync_etf_har	0.003450	0.000703
TLT	0.001222	sync_etf_har	0.001814	0.000593
GLD	0.003178	etf_har	0.003490	0.000312
XLV	0.002747	sync_etf_har	0.003038	0.000292
QQQ	-0.000219	har_only	-0.000219	0.000000
XLK	0.002747	har_only	0.002747	0.000000

Table 2: RMSE summary. HAR-only remains the best RMSE model across all ETFs after rounding, implying that the forecast gains arise through ranking quality rather than point-forecast accuracy.

ETF	HAR-only RMSE	Best RMSE model
GLD	0.029410	har_only
IWM	0.028911	har_only
QQQ	0.027928	har_only
SPY	0.028910	har_only
TLT	0.028915	har_only
XLE	0.028910	har_only
XLF	0.028910	har_only
XLK	0.028910	har_only
XLV	0.028910	har_only
XLY	0.028910	har_only

aligned stock universes. Broad ETFs remain closer to the market-wide benchmark, while sector ETFs produce more distinct high-synchronization calendars. This validates the extension as a meaningful diagnostic, even though the full-market benchmark remains the cleanest central specification.

Finally, the out-of-sample and portfolio evidence suggest that ETF information improves cross-sectional ranking performance and gross portfolio quality, but that turnover remains the main obstacle to net profitability. More broadly, the paper's central message is simple: synchronization changes whether ETF predictability exists, what sign it takes, and when it appears.

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Table 3: Gross portfolio summary by ETF. The final column reports the improvement in gross Sharpe relative to the HAR-only baseline.

ETF	HAR-only gross Sharpe	Best gross model	Best gross Sharpe	Gain vs. HAR
XLE	0.013354	sync_etf_har	0.358866	0.345512
XLF	0.013354	matched_sync_etf_har	0.269985	0.256631
SPY	0.013354	matched_sync_etf_har	0.240203	0.226849
IWM	-0.051004	sync_etf_har	0.153886	0.204890
XLY	0.013354	matched_sync_etf_har	0.203537	0.190183
XLV	0.013354	sync_etf_har	0.131129	0.117774
TLT	-0.197952	sync_etf_har	-0.101132	0.096820
GLD	-0.064183	sync_etf_har	-0.008626	0.055557
XLK	0.013354	matched_sync_etf_har	0.031670	0.018316
QQQ	-0.474033	har_only	-0.474033	0.000000

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